

# Federated and Quantized-Based Intrusion Detection for Embedded On-Board Automotive Networks

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**Abstract**—Intelligent vehicles rely on hybrid On-Board Automotive Networks (OANs), such as Time-Sensitive Networking (TSN) and Controller Area Networks (CAN), to support autonomous driving and ADAS. Securing these networks is challenging due to resource constraints, heterogeneous protocols, and evolving attacks. We propose an embedded-optimized, privacy-preserving collaborative IDS for hybrid OANs. Leveraging Cloud Vehicle Functions (CVFs), the framework enables robust model training across vehicles without sharing raw data. To efficiently process multi-protocol traffic, we introduce the Network Traffic Temporal Encoder (NTTE), which transforms network packets into temporal traffic images for anomaly detection. These NTTE representations are analyzed by TCNnet, a lightweight neural network designed for efficient embedded deployment. To make TCNnet suitable for resource-constrained embedded systems, we propose post-training quantization, significantly reducing its computational and storage demands. Evaluated on real traffic, the quantized TCNnet maintains over 99% detection accuracy, while reducing model size by over 91% and memory footprint by 51%, enabling practical deployment on resource-constrained ECUs. To link IDS outputs to real-world transportation outcomes, we develop the Transportation Impact Modeling Module (TIMM), which maps network-level detection performance to vehicle-level safety metrics (collision likelihood and Safety Message Delivery Ratio, SMDR) and fleet-level traffic indicators (Vehicle Flow Efficiency, VFE). TIMM simulations across Urban, Sub-urban, and Highway scenarios show the IDS reduces expected collisions by up to 97.9%, while maintaining high traffic flow efficiency ( $VFE = 0.949\text{--}0.995$ ) and message reliability ( $SMDR = 0.988\text{--}0.994$ ). These results demonstrate that the proposed IDS not only excels in network-level detection but also directly enhances vehicular safety and traffic performance, confirming its practical applicability in Intelligent Transportation Systems.

**Index Terms**—Intrusion detection system, collaborative learning, intra-vehicular network, cloud vehicle.

## I. INTRODUCTION

THE rise in automation levels (LoA) in intelligent vehicles, as defined by SAE J3016 [1], has increased the number

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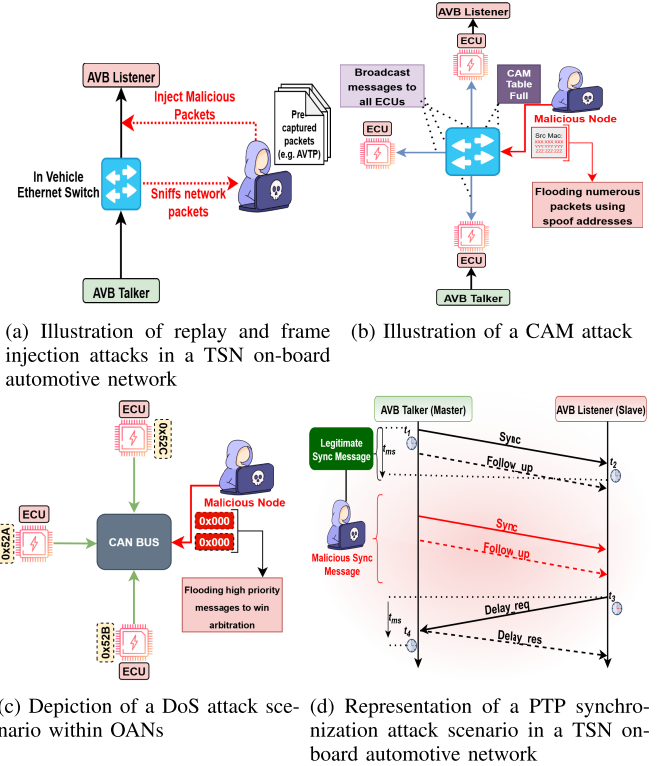


Fig. 1. Illustrations of various attack scenarios in hybrid OANs.

of electronic control units (ECUs) supporting ADAS and autonomous driving. This growth places higher demands on on-board automotive networks (OANs), especially as high-definition sensors strain the bandwidth limits of legacy IVN technologies such as MOST, CAN, and FlexRay [2]. Time-Sensitive Networking (TSN) addresses these limitations by providing bounded latency, guaranteed bandwidth, and precise time synchronization [3]. For instance, IEEE 1722–2016 ensures reliable audio-video transport through AVTP [4], while gPTP enables synchronized transmissions across TSN-based OANs. TSN thus represents a key advancement for modern OANs and is expected to coexist with established technologies like CAN [5]. However, integrating these technologies also introduces security risks. Both TSN and CAN lack strong security mechanisms, making them vulnerable to cyber threats [7], as shown in Fig. 1, including unauthorized access and sensor data manipulation [8]. Such attacks can directly compromise vehicle safety: manipulated AVTP packets may misinform ADAS and trigger hazardous decisions. Fig. 2 shows how an attacker can sniff and compromise AVTP traffic at the talker, inject forged packets to the listener, and

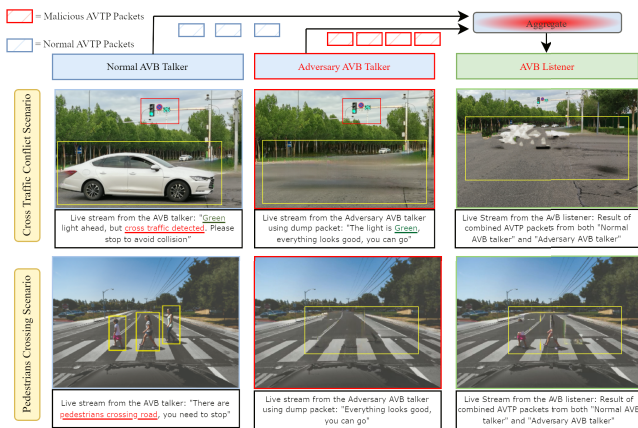


Fig. 2. Illustrating the impact of malicious AVTP packets from a compromised AVB talker on traffic safety communication, potentially causing ADAS to make incorrect decisions and leading to dangerous vehicle maneuvers in cross-traffic and pedestrian scenarios [6].

mislead ADAS into unsafe actions, such as misjudging traffic or failing to avoid collisions. Traditional security measures like encryption and authentication introduce overhead that conflicts with the strict timing requirements of OAN, making real-time security challenging. Intrusion Detection Systems (IDS) provide an alternative by monitoring network behavior without modifying existing nodes [9]. Recently, ML-based IDS have gained traction for detecting complex attacks in OANs [10].

Although ML-based IDS solutions show promise, they face key challenges in intelligent vehicles. Resource constraints limit the deployment of computationally heavy models, while rapidly evolving attacks demand adaptive detection. The heterogeneity of OAN protocols (e.g., AVTP, CAN, gPTP) further complicates feature extraction due to protocol-specific attack behaviors. A critical gap also persists in system evaluation: despite high reported network-level accuracy, the real impact on transportation safety and traffic flow is rarely quantified. Existing IDS frameworks lack a formal methodology to translate network metrics (e.g., TPR) into vehicle-level safety outcomes (e.g., collision probability) and fleet-level efficiency, which are central to ITS. These challenges motivate a collaborative approach in which edge vehicles collectively extract, learn, and share attack patterns in a privacy-preserving manner, allowing them to pool computational resources and adapt to emerging threats without overloading individual nodes. To link network security with transportation safety, we develop the Transportation Impact Modeling Module (TIMM), a framework that formally maps IDS performance metrics (detection latency, TPR, FPR) to ITS-level outcomes. TIMM integrates a discrete-time vehicle dynamics model, a probabilistic attack-detection timing model, and aggregation rules to quantify how IDS behavior influences collision likelihood, safety-message reliability, and traffic-flow efficiency. The main contributions of our work are as follows:

- We propose a privacy-preserving collaborative learning framework with adaptive momentum-aware aggregation, leveraging the Cloud Vehicle Function (CVF) [11] to enable vehicles to collaboratively learn attack patterns and improve model robustness under heterogeneous traffic conditions.

- We develop the TCNnet, a lightweight convolutional model designed for efficient intrusion detection in network traffic. TCNnet extracts spatial features using convolutional layers while leveraging post-training quantization to optimize memory usage and computational efficiency. This makes it particularly suitable for deployment on resource-constrained edge devices.
- We introduce the Network Traffic Temporal Encoder (NTTE), which processes packets from multiple protocols (e.g., AVTP, CAN, gPTP), generating network traffic images that capture temporal anomalies, enhancing the detection of cross-protocol attacks.
- We design the TIMM to rigorously evaluate the real-world transportation impact of the proposed IDS. TIMM translates network-level detection performance into vehicle-level safety metrics (e.g., collision likelihood) and fleet-level traffic efficiency indicators (e.g., Vehicle Flow Efficiency), thereby directly linking cybersecurity performance to ITS safety and operational goals.

## II. RELATED WORK

The authors in [12] designed an approach to derive two distinctive features of CAN bus communication: the time interval between successive frames (interframe space or IFS) and the counter-related values embedded in the data payload. These features form the basis of their decision tree-driven IDS. To assess their effectiveness, the authors conducted evaluations using well-known ensemble learning algorithms, specifically Random Forest (RF) and XGBoost classifiers. In [13], the authors introduced CANShield, a DL-driven framework designed for signal-level monitoring of the CAN bus. Their proposed architecture comprises three core components: (a) a data preprocessing module that transforms the high-dimensional CAN signal stream into structured time-series data suitable for DL; (b) a data analysis module employing multiple deep autoencoders, each operating at distinct temporal resolutions to capture patterns across varying time scales; and (c) an attack detection module that aggregates the outputs using an ensemble-based decision strategy to identify malicious activity. The authors in [14] proposed an ML-based IDS to detect attacks in intra-vehicle networks, especially in CAN. Their proposed IDS utilizes Recurrence Plot (RP) to exploit the temporal properties of CAN messages as well as intra- and inter-message dependencies to create high-level representations of CAN messages that are transmitted on the bus. These representations are then fed into a custom neural network that is trained to identify new intrusions. In [15], the authors proposed a multi-stage IDS, where in the first stage, the authors utilize an artificial neural network to detect known attacks, while the second stage comprises an LSTM autoencoder for new and unseen attack detection. Further, the authors used a hierarchical federated learning environment to preserve data privacy, analyze diverse driving behavior, and update the model with the latest patterns of attacks.

Furthermore, the authors in [16] proposed a multi-stage IDS based on DL for the detection and classification of cyberattacks in Automotive Ethernet Networks (AENs). In the first stage, the authors employed RF for cyberattacks detection, while the authors utilized a Pruned CNN in the

second stage to minimize the false positive rate of their IDS during the classification of various attacks. Similarly, the authors in [17] designed an IDS named “AERO” for in-vehicle networks. Their proposed “AERO” IDS comprises three modules, i.e., a feature extractor for the construction of three multimodal features, a neural network to process the extracted features, and an online anomaly detector for the calculation of the outlier scores in real time. In [18], the authors propose an evaluation framework designed to support reproducible, consistent, and efficient benchmarking of detection algorithms. Their proposed framework utilizes a simulation toolchain that offers customizable network topologies, traffic patterns, anomaly injections, attack scenarios, and detection mechanisms. To illustrate its capabilities, the authors evaluate their Network Anomaly Detection Systems (NADSS) within a detailed in-vehicle communication environment, simulating realistic traffic anomalies. Moreover, the authors of [19] introduce an IDS to detect the audio-video transport protocol stream injection attacks in AENs. Their proposed IDS is based on feature generation and CNN. Further, the authors built a physical BroadR-Reach-based testbed and captured real audio-video transport protocol packets to evaluate their proposed IDS.

While existing IDS solutions for in-vehicle networks have shown promise, they face key limitations. Many are too resource-intensive for embedded environments, lack support for hybrid protocols like CAN and TSN, and do not address privacy concerns in collaborative settings. Moreover, current models often struggle to adapt to evolving or zero-day attacks due to their reliance on static training data. To address these challenges, our proposed IDS integrates a lightweight, quantized Tiny CNN (TCNnet) for efficient deployment, a Network Traffic Temporal Encoder (NTTE) for cross-protocol anomaly detection, and a privacy-preserving collaborative learning framework using Cloud Vehicle Functions (CVFs).

### III. PROPOSED FRAMEWORK

In this section, we first provide the details of the encoding mechanism, followed by the architecture of a deep learning-based intrusion detector. Finally, we present the proposed privacy-preserved collaborative IDS.

#### A. Network Traffic Temporal Encoder (NTTE)

The first module of our proposed framework is the Network Traffic Temporal Encoder (NTTE). This module processes traffic packets from various protocols received from network nodes via port mirroring (e.g., BroadR-Reach) and encodes these packets into network images. NTTE is designed to accommodate heterogeneous protocol formats, including CAN, TSN, and IP-based traffic, in order to standardize packet structures through protocol-independent byte extraction and alignment. These images serve as feature representations and are used as input to the DL-based intrusion detector. Let  $\Xi^{[k]}$  denote a one-dimensional column vector representing the  $k$ -th traffic packet. Formally, it is defined as:

$$\Xi^{[k]} = \begin{pmatrix} \Psi_1^{[k]} \\ \Psi_2^{[k]} \\ \vdots \\ \Psi_n^{[k]} \end{pmatrix}, \quad (1)$$

where each  $\Psi_i^{[k]}$  represents a payload byte of the traffic packet  $\Xi^{[k]}$  and satisfies the condition  $\Psi_i^{[k]} \in \mathbb{N}$  with  $0 \leq \Psi_i^{[k]} < 256$ . The total number of bytes in the traffic packet is denoted by  $n$ . The size of  $\Xi^{[k]}$  may vary depending on the type of protocol.

To analyze temporal dependencies between network packets, we group them using an overlapping sequence window of length  $\Upsilon \in \mathbb{N}$ , where  $\Upsilon \geq 1$ . The collected packets within each sequence window are expressed as:

$$\Xi^{[\Upsilon]} = (\Xi^{[k]} \ \Xi^{[k+1]} \ \Xi^{[k+2]} \ \dots \ \Xi^{[k+(\Upsilon-1)]}), \quad (2)$$

where  $\Xi^{[\Upsilon]}$  is a structured matrix aggregating multiple packets.

To advance the window along the sequence of packets, we introduce the stride parameter  $\tau \in \mathbb{N}$ , where  $\tau \geq 1$ . This parameter controls the step size by which the window shifts, thereby influencing the amount of temporal context preserved from the previous window.

For clarity, the aggregated packets along with each element  $\Psi_i^{[k+j]}$  are structured in the following matrix representation:

$$\Xi^{[\Upsilon]} = \begin{pmatrix} \Psi_1^{[k]} & \Psi_1^{[k+1]} & \Psi_1^{[k+2]} & \dots & \Psi_1^{[k+(\Upsilon-1)]} \\ \Psi_2^{[k]} & \Psi_2^{[k+1]} & \Psi_2^{[k+2]} & \dots & \Psi_2^{[k+(\Upsilon-1)]} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Psi_n^{[k]} & \Psi_n^{[k+1]} & \Psi_n^{[k+2]} & \dots & \Psi_n^{[k+(\Upsilon-1)]} \end{pmatrix}, \quad (3)$$

where each column corresponds to a single traffic packet, and each row represents a specific byte position across the aggregated packets.

After the aggregation of packets, the next step is to extract a fixed number of bytes,  $s$ , from each packet in  $\Xi^{[\Upsilon]}$ . Since OANs utilize multiple protocols with varying traffic image sizes, a standardized packet length is required to ensure consistency. To achieve this, we apply mean padding instead of zero padding, as it better preserves statistical characteristics of the data. The resulting structured representation, denoted as  $\Xi^{[\text{SPS}]}$  (Standardized traffic image size), is given by:

$$\Xi^{[\text{SPS}]} = \begin{pmatrix} \Psi_1^{[k]} & \Psi_1^{[k+1]} & \Psi_1^{[k+2]} & \dots & \Psi_1^{[k+(\Upsilon-1)]} \\ \Psi_2^{[k]} & \Psi_2^{[k+1]} & \Psi_2^{[k+2]} & \dots & \Psi_2^{[k+(\Upsilon-1)]} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Psi_s^{[k]} & \Psi_s^{[k+1]} & \Psi_s^{[k+2]} & \dots & \Psi_s^{[k+(\Upsilon-1)]} \end{pmatrix}, \quad (4)$$

where each element  $\Psi_m^{[k+j]}$  of  $\Xi^{[\text{SPS}]}$  is defined as:

$$\Psi_m^{[k+j]} = \begin{cases} \Psi_i^{[k+j]}, & \text{if } i \leq n(k+j), \\ \mu, & \text{if } i > n(k+j). \end{cases} \quad (5)$$

where,  $\Psi_i^{[k+j]}$  represents the  $i$ -th byte of the  $\Xi^{[k]}$ -th packet.

To capture the temporal changes between consecutive packets, we compute the modulus of the difference between two consecutive packets  $\Xi^{[k-1]}$  and  $\Xi^{[k]}$ . This difference is defined as:

$$\Delta\Xi^{[k]} \equiv (\Xi^{[k]} - \Xi^{[k-1]}) \bmod 2^8. \quad (6)$$

The rate of change  $\Delta$  can be expanded to all packets within the window size  $\Upsilon$ , and formally defined as:

$$\Delta\Xi^{[\Upsilon]} = \begin{pmatrix} \Delta\Xi^{[k]} & \Delta\Xi^{[k+1]} & \Delta\Xi^{[k+2]} \\ \dots & \Delta\Xi^{[k+(\Upsilon-1)]} \end{pmatrix}, \quad (7)$$

where  $\Delta\Xi^{[k]}$  represents the state of change between the  $\Xi^{[k]}$ -th and  $\Xi^{[k-1]}$ -th packets.



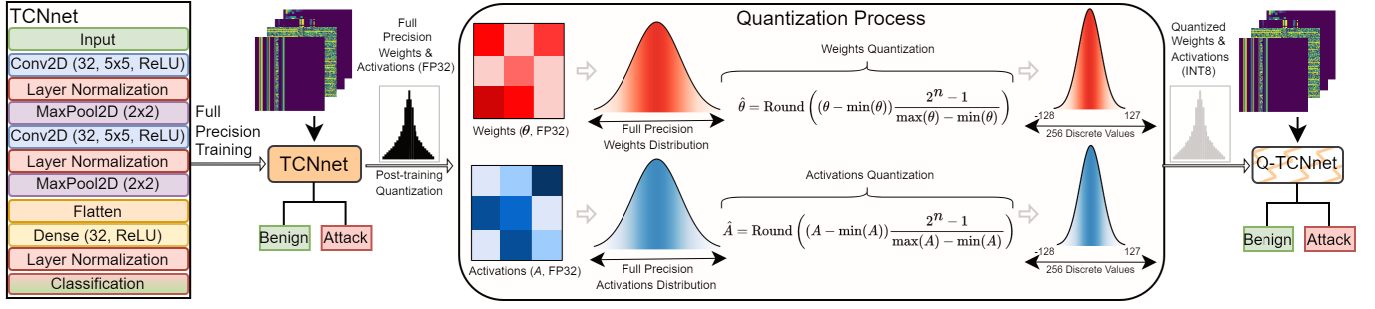


Fig. 3. Illustration of proposed TCNnet with post-quantization process.

The matrix of changes for all elements in each packet within a specified window size for SPS can be formalized as follows:

$$\Delta \Xi^{[SPS]} = \begin{pmatrix} \Delta \Psi_1^{[k]} & \Delta \Psi_1^{[k+1]} & \Delta \Psi_1^{[k+2]} & \dots & \Delta \Psi_1^{[k+(Y-1)]} \\ \Delta \Psi_2^{[k]} & \Delta \Psi_2^{[k+1]} & \Delta \Psi_2^{[k+2]} & \dots & \Delta \Psi_2^{[k+(Y-1)]} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Delta \Psi_s^{[k]} & \Delta \Psi_s^{[k+1]} & \Delta \Psi_s^{[k+2]} & \dots & \Delta \Psi_s^{[k+(Y-1)]} \end{pmatrix} \quad (8)$$

Finally, to better analyze the data within packets, each element  $\Delta \Psi_m^{[k+j]}$  (byte) of  $\Delta \Xi^{[SPS]}$  is further divided into 4-bit segments (e.g. nibbles). This process is formalized as:

$$\Delta \tilde{\Psi}_m^{[k+j]} = \begin{cases} \left\lfloor \frac{\Delta \Psi_m^{[k+j]}}{2^4} \right\rfloor, & \text{if } m \text{ is odd,} \\ \Delta \Psi_m^{[k+j]} \& 0 \times 0F, & \text{if } m \text{ is even.} \end{cases} \quad (9)$$

The resulting detailed matrix  $\Delta \tilde{\Xi}^{[SPS]}$  is represented as:

$$\Delta \tilde{\Xi}^{[SPS]} = \begin{pmatrix} \Delta \tilde{\Psi}_1^{[k]} & \Delta \tilde{\Psi}_1^{[k+1]} & \Delta \tilde{\Psi}_1^{[k+2]} & \dots & \Delta \tilde{\Psi}_1^{[k+(Y-1)]} \\ \Delta \tilde{\Psi}_2^{[k]} & \Delta \tilde{\Psi}_2^{[k+1]} & \Delta \tilde{\Psi}_2^{[k+2]} & \dots & \Delta \tilde{\Psi}_2^{[k+(Y-1)]} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \Delta \tilde{\Psi}_s^{[k]} & \Delta \tilde{\Psi}_s^{[k+1]} & \Delta \tilde{\Psi}_s^{[k+2]} & \dots & \Delta \tilde{\Psi}_s^{[k+(Y-1)]} \end{pmatrix}. \quad (10)$$

The intrusion detector takes  $\Delta \tilde{\Xi}^{[SPS]}$  as its input. The core purpose of the detector is to recognize deviations between normal and malicious packets by analyzing the pattern of packet changes. To achieve this, each input matrix  $\Delta \Xi^{[SPS]}$  is labeled based on its corresponding network activity, where normal traffic is assigned a benign label, and attack patterns are categorized accordingly. The details of the labeling process for both binary and multi-class traffic classification are provided in Subsection III-C.

### B. Intrusion Detector

Our proposed intrusion detector, *TCNnet*, is a lightweight Tiny Convolutional Neural Network tailored for network-traffic intrusion detection. It extracts spatial features using stacked convolutional layers, followed by layer normalization and max pooling to preserve salient patterns while improving generalization. The complete workflow, including post-training quantization, is shown in Fig. 3.

1) *Tiny Convolutional Neural Network (TCNnet)*: TCNnet consists of convolutional, normalization, pooling, and fully connected layers, all quantized post-training using TensorFlow Lite to reduce model size and inference latency. The core operations for each layer are summarized below.

a) *Convolutional layers with quantization*: The convolutional feature extraction follows:

$$h^{[l+1]}(x) = AF \left( \sum_{j=0}^{M-1} \sum_{i=0}^{N-1} \theta_{ij}^{[l]} \times x_{ij}^{[l]} + Bias^{[l]} \right), \quad (11)$$

with weights and activations quantized post-training:

$$\hat{\theta}_{ij}^{[l]} = \left\lfloor \frac{\theta_{ij}^{[l]} - \min(\theta^{[l]})}{\Delta^{[l]}} \right\rfloor, \quad \hat{A}^{[l]} = \left\lfloor \frac{A^{[l]} - \min(A^{[l]})}{\Delta_A^{[l]}} \right\rfloor \quad (12)$$

with corresponding quantization steps:

$$\Delta^{[l]} = \frac{\theta_{\max}^{[l]} - \theta_{\min}^{[l]}}{2^n - 1}, \quad \Delta_A^{[l]} = \frac{A_{\max}^{[l]} - A_{\min}^{[l]}}{2^n - 1}. \quad (13)$$

b) *Layer normalization and max pooling*: Layer normalization stabilizes activations:

$$\mathcal{X} = \frac{X - \mu_X}{\sqrt{\sigma_X^2 + \epsilon}}, \quad Z^{norm} = \gamma \mathcal{X} + \beta, \quad (14)$$

followed by max pooling for spatial downsampling.

2) *Fully Connected Layer With Quantization*: Fully connected operations follow:

$$A = \theta X + B, \quad (15)$$

with quantized weights and activations:

$$\hat{\theta} = \left\lfloor \frac{\theta - \min(\theta)}{\Delta_{\theta}} \right\rfloor, \quad \hat{A} = \left\lfloor \frac{A - \min(A)}{\Delta_A} \right\rfloor, \quad (16)$$

where

$$\Delta_{\theta} = \frac{\theta_{\max} - \theta_{\min}}{2^n - 1}, \quad \Delta_A = \frac{A_{\max} - A_{\min}}{2^n - 1}. \quad (17)$$

A ReLU activation is applied afterward:

$$A' = \max(0, \hat{A}). \quad (18)$$

3) *Output Layer and Softmax*: The final logits are computed as:

$$Y_f = \hat{\theta}_f \hat{A}' + B_f, \quad (19)$$

with quantized output weights:

$$\hat{\theta}_f = \left\lfloor \frac{\theta_f - \theta_{f,\min}}{\Delta_{\theta_f}} \right\rfloor, \quad \Delta_{\theta_f} = \frac{\theta_{f,\max} - \theta_{f,\min}}{2^n - 1}. \quad (20)$$

Softmax yields class probabilities:

$$P(\hat{Y})_i = \frac{e^{Y_f^i}}{\sum_{j=1}^C e^{Y_f^j}}. \quad (21)$$



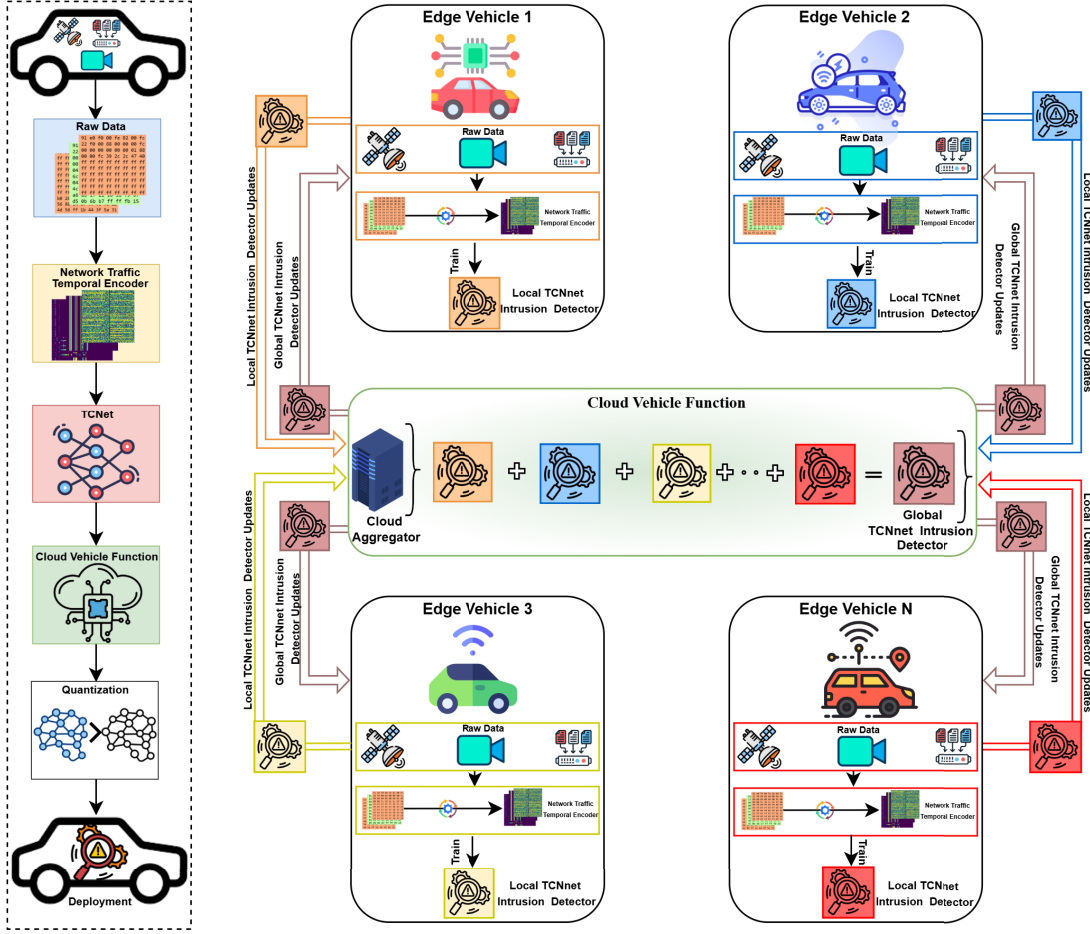


Fig. 4. Illustration of proposed collaborative learning framework with adaptive momentum-aware aggregation strategy.

4) *Loss Function*: Training minimizes Categorical Cross-Entropy:

$$L = - \sum_{i=1}^C Y_i \log(P(\hat{Y}_i)). \quad (22)$$

Overall, TCNnet applies quantization to all major layers (convolutional, normalization, and fully connected), significantly reducing memory and computation while preserving detection accuracy, making it suitable for deployment in resource-constrained environments.

### C. Privacy-Preserved Collaborative IDS

The next module of the framework is the collaborative learning stage, illustrated in Fig. 4. The CVF aggregator initializes the TCNnet detector and broadcasts it to all edge vehicles. Each vehicle preprocesses its local traffic using NTTE (Section III-A), producing traffic images for TCNnet. These images are then labeled using two criteria: (i) binary (normal vs. attack) and (ii) multi-class (specific attack types), formally defined in Algorithms 1 and 2. The labeled images are used to train TCNnet locally, after which each vehicle sends its model updates to the aggregator. The aggregator fuses

**Algorithm 1** Labeling Criteria for Binary Traffic (AEID)  $\Upsilon$  Is the Window Size,  $\Delta\Xi^{[SPS]}$  Is a Group of Malicious Packets, and  $\Xi^{[\Upsilon]}$  Is a Stream of Raw Packets

```

1: procedure LABELPACKETS ( $\Xi^{[\Upsilon]}$ ,  $\Upsilon$ ,  $\Delta\Xi^{[SPS]}$ )
2:   Initialize an empty set to store labels for  $\Upsilon$ :  $\Gamma \leftarrow \{\}$ 
3:   for each window  $w_i$  of size  $\Upsilon \in \Xi^{[\Upsilon]} = [0, \dots, n-1]$  do
4:     Check if  $w_i$  contains any packet in  $\Delta\Xi^{[SPS]}$ 
5:     if  $w_i \cap \Delta\Xi^{[SPS]} \neq \{\}$  then
6:       Assign binary label:  $\Gamma_{w_i} \leftarrow [0, 1] \triangleright [\text{Normal}, \text{Attack}]$ 
7:     else
8:       Assign binary label:  $\Gamma_{w_i} \leftarrow [1, 0] \triangleright [\text{Normal}, \text{Attack}]$ 
9:     end if
10:    Append the label to the set of labels:  $\Gamma \leftarrow \Gamma \cup \{(w_i, \Gamma_{w_i})\}$ 
11:  end for
12:  return  $\Gamma$ 
13: end procedure

```

the received updates and redistributes the refined global model. This process repeats iteratively until convergence or until a predefined performance threshold is met.

Let's assume a vehicular network consisting of a set of edge vehicles denoted as  $D = \{1, 2, 3, \dots, n\}$ , where each vehicle  $d_i$

**Algorithm 2** Labeling Criteria for Multi-Class Traffic (TOW)  
 $\Upsilon$  Is the Window Size,  $\mathbb{A}$  Is a Set of Attack Types, and  $\Xi^{[\Upsilon]}$   
 Is a Stream of Raw Packets

```

1: procedure LABELPACKETS ( $\Xi^{[\Upsilon]}$ ,  $\Upsilon$ ,  $\mathbb{A}$ )
2:   Initialize an empty set to store labels for  $\Upsilon$ :  $\Gamma \leftarrow \{\}$ 
3:   for each window  $w_i$  of size  $\Upsilon \in \Xi^{[\Upsilon]} = [0, \dots, n-1]$  do
4:     Initialize one-hot vector:  $\Gamma_{w_i} \leftarrow [1, 0, \dots, 0]$   $\triangleright$ Default is Normal
5:     for each attack type  $a_j$  in  $\mathbb{A}$  do
6:       if  $w_i \cap a_j \neq \{\}$  then
7:         Set attack label:  $\Gamma_{w_i}[j] \leftarrow 1$ ,  $\Gamma_{w_i}[0] \leftarrow 0$ 
8:       break
9:     end if
10:  end for
11:  Append the label to the set of labels:  $\Gamma \leftarrow \Gamma \cup \{(w_i, \Gamma_{w_i})\}$ 
12: end for
13: return  $\Gamma$ 
14: end procedure

```

holds a set of traffic packets represented as:

$$\Xi^{[d_i]} = \{\Xi_1^{[d_i]}, \Xi_2^{[d_i]}, \Xi_3^{[d_i]}, \dots, \Xi_n^{[d_i]}\} \quad (23)$$

These traffic flows are processed by NTTE to generate traffic images for intrusion detection. Each image is represented by a feature matrix  $\Xi_j^{[d_i]}$  that encodes the key characteristics distinguishing normal and attack behaviors. Thus, each vehicle  $d_i$  maintains a dataset of labeled traffic images defined as:

$$\mathbb{D}^{[d_i]} = \{(\Xi_j^{[d_i]}, \Gamma_j^{[d_i]}) \mid j = 1, 2, \dots, M^{[d_i]}\}, \quad (24)$$

where  $\Xi_j^{[d_i]}$  is the feature matrix corresponding to the  $j$ -th traffic image, and  $\Gamma_j^{[d_i]}$  represents its attack type label. The total number of traffic images stored in each vehicle is given by  $M^{[d_i]} = |\mathbb{D}^{[d_i]}|$ .

For a detection task, the attack label  $\Gamma_j^{[d_i]}$  is represented as a one-hot encoded vector:

$$\Gamma_j^{[d_i]} = [\Gamma_{[j,0]}^{[d_i]}, \Gamma_{[j,1]}^{[d_i]}, \dots, \Gamma_{[j,\Omega-1]}^{[d_i]}], \quad (25)$$

where  $\Omega$  is the total number of attack types.

DL-based intrusion detectors are trained by minimizing an objective function, where the overall loss aggregates the errors across all training samples. This optimization process is expressed as follows:

$$\min_{\theta \in \mathbb{R}^d} F(\theta), \quad (26)$$

where

$$F(\theta) = \frac{1}{M} \sum_{i=1}^M \mathcal{L}oss(\Xi_i^{[d_i]}, \Gamma_i^{[d_i]}; \theta), \quad (27)$$

and  $\mathcal{L}oss(\Xi_i^{[d_i]}, \Gamma_i^{[d_i]}; \theta)$  is the loss function computed for the feature matrix  $\Xi_i^{[d_i]}$  and its corresponding attack label  $\Gamma_i^{[d_i]}$  with respect to the TCNnet parameters  $\theta$ .

A gradient update involves computing the gradient of  $F(\theta)$ , which can be expressed as:

$$\mathfrak{d}^{mean}(\theta) = \nabla F(\theta) = \frac{1}{M} \sum_{i=1}^M \mathfrak{d}_i(\theta), \quad (28)$$

where

$$\mathfrak{d}_i(\theta) = \frac{\partial \mathcal{L}oss(\Xi_i^{[d_i]}, \Gamma_i^{[d_i]}; \theta)}{\partial \theta}. \quad (29)$$

Instead of using the entire traffic image dataset in each optimization step, a subset of traffic images, denoted by  $\mathcal{S} \subset \{1, 2, \dots, n\}$ , is randomly chosen at each iteration. This results in a modified loss function that focuses only on the selected subset is defined as:

$$\min_{\theta \in \mathbb{R}^d} F_{\mathcal{S}}(\theta), \quad (30)$$

where

$$F_{\mathcal{S}}(\theta) = \frac{1}{n_{\mathcal{S}}} \sum_{i \in \mathcal{S}} \mathcal{L}oss(\Xi_i^{[d_i]}, \Gamma_i^{[d_i]}; \theta), \quad (31)$$

with  $n_{\mathcal{S}}$  representing the total number of selected traffic images,  $n_{\mathcal{S}} = |\mathcal{S}|$ .

We also compute the gradient  $\mathfrak{d}$  only for the selected traffic images. This gradient for the subset can be represented as:

$$\mathfrak{d}_{\mathcal{S}}(\theta) = \nabla F_{\mathcal{S}}(\theta) = \frac{1}{n_{\mathcal{S}}} \sum_{i \in \mathcal{S}} \mathfrak{d}_i(\theta), \quad (32)$$

The earlier loss function applies to a single vehicle  $d_i$ . In the collaborative setting, multiple vehicles participate in training; thus, the loss must be extended to a set of  $n$  contributing vehicles, expressed as:

$$F(\theta) = \arg \min_{\theta \in \mathbb{R}^d} \sum_{i=1}^n \frac{D^{[d_i]}}{D} F_{d_i}(\theta), \quad (33)$$

where

$$F_{d_i}(\theta) = \frac{1}{D^{[d_i]}} \sum_{j \in M^{[d_i]}} \mathcal{L}oss(\Xi_j^{[d_i]}, \Gamma_j^{[d_i]}; \theta). \quad (34)$$

In this revised formulation, rather than computing the loss function by averaging over a local set of  $M$  traffic images, expressed as  $\frac{1}{M} \sum_{i=1}^M F_i(\theta)$ , we first compute the individual loss function for each edge vehicle  $d_i$ . The local objective function for a given vehicle  $d_i$  is defined as:

$$F_{d_i}(\theta) = \frac{1}{|\mathbb{D}^{[d_i]}|} \sum_{j=1}^{M^{[d_i]}} \mathcal{L}oss(\Xi_j^{[d_i]}, \Gamma_j^{[d_i]}; \theta). \quad (35)$$

Following this, a weighted aggregation is performed over all participating  $n$  vehicles, where the weight assigned to each vehicle is proportional to the number of traffic images  $|\mathbb{D}^{[d_i]}|$  it possesses.

For each vehicle  $d_i$ , an input sample  $\Xi_j^{[d_i]}$  is transformed into a network traffic image, denoted as  $\Delta \Xi^{[SPS]} \in \mathbb{R}^{(\Upsilon-1) \times s\text{-bytes}}$ . The corresponding output label  $\Gamma_j^{[d_i]}$  is represented as a one-hot encoded vector in  $\mathbb{R}^{\Omega}$ , where  $\Omega$  is the total number of attack categories. Each vehicle maintains a local dataset  $\mathbb{D}^{[d_i]}$ , consisting of  $M^{[d_i]}$  labeled traffic samples. To minimize its empirical risk, each vehicle independently optimizes its local objective function. The overall global empirical risk, considering all participating vehicles, is expressed as:

$$F(\theta_1, \dots, \theta_n) = \arg \min_{\theta_i \in \mathbb{R}^{|\mathbb{D}^{[d_i]}|}} \sum_{i=1}^n \frac{|\mathbb{D}^{[d_i]}|}{|\mathbb{D}|} F_{d_i}(\theta). \quad (36)$$

Before the commencement of the initial communication round in our collaborative learning paradigm, an aggregator

at CVF initializes the TCNnet model and disseminates it to all selected edge vehicles in the network, formally defined as:

$$\theta_{d_i} \leftarrow \theta^{[\text{global}]}, \quad (37)$$

where  $\theta^{[\text{global}]}$  represents the global TCNnet parameters, while  $\theta_{d_i}$  refers to the local TCNnet parameters assigned to each edge vehicle  $d_i$ . Beyond maintaining individual vehicle loss function as formulated in Eq. 35, each vehicle must compute its gradient to facilitate parameter updates. Within this collaborative framework, the local gradient  $\nabla F_{d_i}$  is computed over the vehicle's private dataset as:

$$\nabla F_{d_i}(\theta) = \frac{1}{|\mathbb{D}[d_i]|} \sum_j \nabla F_j(\theta). \quad (38)$$

Subsequently, the global gradient  $\nabla F^{[\text{global}]}$  across all participating vehicles is derived using an adaptive momentum-aware aggregation strategy:

$$\nabla F_t^{[\text{global}]} = \sum_{i=1}^n \frac{|\mathbb{D}[d_i]|}{|\mathbb{D}|} \left[ \beta \cdot \nabla F_{t-1}^{[\text{global}]} + (1 - \beta) \cdot \nabla F_{d_i}^{(t)}(\theta) \right], \quad (39)$$

where  $\beta \in [0, 1)$  is the momentum coefficient and  $t$  is the current communication round. This aggregation balances past and recent gradients, improving stability and convergence under non-IID, heterogeneous vehicular data, and enabling a more robust global IDS model.

These steps are repeated for a fixed number of iterations. Algorithm 3 summarizes the full collaborative IDS framework, integrating NTTE, TCNnet, and the privacy-preserved federated learning process.

#### D. Transportation Impact Modeling Module (TIMM)

To formally connect network-level intrusion detection outputs to vehicle-level safety and traffic-flow indicators, we introduce a Transportation Impact Modeling Module (TIMM). TIMM consists of (i) a discrete-time longitudinal vehicle dynamics model with perception channel modeling that captures corrupted/filtered V2X/CAM information; (ii) a probabilistic attack-detection timing model that uses IDS performance statistics (TPR, FPR,  $t_{det}$ ); and (iii) aggregation rules that map vehicle-level events to network-level flow metrics. We adopt discrete-time notation with sampling interval  $\Delta t > 0$ . Let time indices be  $k \in \mathbb{N}$  and physical time  $t = k\Delta t$ .

1) *Vehicle State and Controller Model*: Consider a platoon/vehicle string of  $N$  vehicles indexed by  $i = 1, \dots, N$ , where vehicle  $i = 1$  is the leader. For each vehicle  $i$  at time step  $k$  define the state vector:

$$x_i[k] = \begin{bmatrix} p_i[k] \\ v_i[k] \end{bmatrix}, \quad (40)$$

with longitudinal position  $p_i$  and speed  $v_i$ . The discrete-time kinematics are:

$$p_i[k+1] = p_i[k] + v_i[k]\Delta t, \quad v_i[k+1] = v_i[k] + a_i[k]\Delta t, \quad (41)$$

where  $a_i[k]$  is the longitudinal acceleration applied by the vehicle controller.

#### Algorithm 3 Embedded-Optimized Intrusion Detection

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```

1: Procedure CLOUD VEHICLE FUNCTION EXECUTES:
2: Initialize global model parameters  $\theta_0^{[\text{global}]}$ , communication round  $C = 0$ 
3: while not converged or communication rounds  $C < N$  do
4:    $M \leftarrow \text{MAX}(K \times V, 1)$   $\triangleright K$  = fraction of vehicles selected per round
5:    $S_t \leftarrow$  random set of  $M$  vehicles
6:   for each vehicle  $[d_i] \in \{0, 1, \dots, S_t - 1\}$  in parallel do
7:      $\theta_{t+1}^{[d_i]} \leftarrow \text{LocalUpdate}(d_i, \theta_t^{[\text{global}]})$   $\triangleright$  Local TCNnet update
8:   end for
9:    $\theta_{t+1} \leftarrow \sum_{i=1}^n \frac{D^{[d_i]}}{D} \theta_{t+1}^{[d_i]}$   $\triangleright$  Aggregate updates: weighted sum, where  $D^{[d_i]}$  is the local traffic image dataset size and  $D$  is the overall size
10:   $C \leftarrow C + 1$   $\triangleright$  Proceed to the next Comm. Round
11: end while
12: Procedure EDGE VEHICLE EXECUTES:
13: Receive TCNnet parameters  $\theta_t^{[\text{global}]}$  from aggregator
14: Network Traffic Processing:
15: * Extract traffic packets  $\Xi^{[d_i]}$ 
16: * Transform packets into network traffic images dataset:
     $\mathbb{D}^{[d_i]} \leftarrow \text{NTTE}(\Xi^{[d_i]})$ 
17: Label images using appropriate algorithm:
18: if TrafficType == Binary then
19:    $\Gamma^{[d_i]} \leftarrow \text{BinClass}(\Xi^{[d_i]})$  (refer to Algorithm 1)
20: else
21:    $\Gamma^{[d_i]} \leftarrow \text{MultiClass}(\Xi^{[d_i]})$  (refer to Algorithm 2)
22: end if
23: TCNnet Model Training:
24: Initialize TCNnet model  $\theta_t^{[d_i]}$  from aggregator
25: Split labeled dataset  $M^v$  into batches of size  $S$ 
26: for each local epoch  $i \in \{0, 1, \dots, E - 1\}$  do
27:   for each mini-batch  $s \in S$  do
28:     Compute gradient:  $g_s^{[d_i]} \leftarrow \nabla \text{Loss}(\theta_t; s)$ 
29:     Local update:  $\theta_{t+1}^{[d_i]} \leftarrow \theta_t^{[d_i]} - \eta \times g_s^{[d_i]}$ 
30:   end for
31: end for
32: Send updated TCNnet  $\theta_{t+1}^{[d_i]}$  to CVF
33: Post-Training Quantization:
34: Optimize TCNnet for embedded deployment:
     $\theta_{t+1}^{[d_i]} \leftarrow \text{Quantize}(\theta_{t+1}^{[d_i]})$ 
35: End Procedure

```

---

We use a *perception-based linear car-following controller* [20] (suitable for ADAS/platoon controllers). Let the controller command be:

$$a_i[k] = K_p (\hat{v}_{i-1 \rightarrow i}[k] - v_i[k]) + K_h (s^* - s_i[k]) \quad (42)$$

where:

- $s_i[k] = p_{i-1}[k] - p_i[k]$  is the gap to the preceding vehicle,
- $s^*$  is the desired spacing (safe headway),
- $K_p, K_h > 0$  are the control gains,



- $\hat{v}_{i-1 \rightarrow i}[k]$  is the perceived speed of the *preceding vehicle* at vehicle  $i$  (this may be derived from received CAM/V2X messages or local sensing.)

*Perception Channel Model:* The perceived speed of the preceding vehicle  $\hat{v}_{i-1 \rightarrow i}[k]$  equals the true speed  $v_{i-1}[k]$  unless an attack has corrupted the information or the IDS has filtered it:

$$\hat{v}_{i-1 \rightarrow i}[k] = \begin{cases} \tilde{v}_{i-1}[k], & \text{if a malicious message is received} \\ & \text{at } k \text{ and not yet mitigated,} \\ v_{i-1}[k], & \text{otherwise.} \end{cases} \quad (43)$$

Here  $\tilde{v}_{i-1}[k]$  is the attacker-specified (spoofed) value. IDS actions (detection + mitigation) can restore correct perception by blocking or flagging corrupted CAMs. False positives can also cause legitimate messages to be dropped, producing delayed perception updates (stale values).

2) *Attack and IDS Timing Model:* Model an attack instance arriving at time  $t_0$ . Let  $t_{\text{det}}$  denote the IDS detection latency for that instance, and let  $F_t(\tau)$  be the Cumulative Distribution Function (CDF) of detection latency. Formally define:

- $t_c$ : critical time window for which corrupted perception can cause an unsafe control action (time-to-act). For a particular follower  $i$ ,  $t_c$  can be approximated by the Time-To-Collision (TTC) threshold or the controller reaction window (see Section V).
- $\mathbb{I}_{\text{harm}} = \mathbf{1}\{t_{\text{det}} \geq t_c\}$ : indicator that detection is too late to prevent harm. More generally, if the latency is random:

$$P(t_{\text{det}} \geq t_c) = 1 - F_t(t_c). \quad (44)$$

Assume the IDS correctly identifies an attack with probability equal to the recall/TPR and falsely flags a benign message with probability FPR. We model the effect of IDS+mitigation as instantaneous upon detection.

Let  $A$  denote an attack event; for a single attack, the probability that the attack remains effective (i.e., causes the vehicle to act on corrupted perception before mitigation) is:

$$P_{\text{eff}} = P(A) \cdot (1 - \text{TPR} \cdot F_t(t_c)), \quad (45)$$

where  $P(A)$  is the probability that the message is manipulated (given prevalence). For point estimates using mean latency  $\bar{t}_{\text{det}}$  and deterministic threshold:

$$P_{\text{eff}} \approx P(A) \cdot (1 - \text{TPR} \cdot \mathbf{1}\{\bar{t}_{\text{det}} < t_c\}). \quad (46)$$

If the attack is effective for time window  $T_{\text{eff}} = \min(t_c, t_{\text{det}})$  (i.e., until detection or until critical harm occurs), the follower may execute a sequence of corrupted control updates during  $\lceil T_{\text{eff}}/\Delta t \rceil$  steps.

3) *Collision Likelihood for a Vehicle Pair:* For a follower  $i$ , define the instantaneous time-to-collision (TTC) under perceived data as:

$$\text{TTC}_i[k] = \frac{s_i[k]}{\max(0, v_i[k] - \hat{v}_{i-1 \rightarrow i}[k]) + \epsilon}, \quad (47)$$

with small  $\epsilon > 0$  to avoid division by zero. A collision is declared at step  $k$  if  $s_i[k] \leq s_{\text{min}}$  (vehicle overlap) or  $\text{TTC}_i[k] < \tau_{\text{crit}}$ . This definition yields a binary collision outcome per realization, while the overall collision likelihood

$P_{\text{coll}|A}$  remains probabilistic, as it depends on the stochastic nature of perception errors and detection latency.

For probabilistic analysis, the conditional probability that an ongoing effective attack leads to a collision within horizon  $H$  is given by:

$$P_{\text{coll}|A} = P(\exists k \in \{k_0, \dots, k_0 + H\} : \text{TTC}_i[k] < \tau_{\text{crit}} \mid A \text{ effective}), \quad (48)$$

which represents the general probabilistic formulation. In practice, we employ two complementary approaches: (i) an analytical conservative bound based on deterministic gap-closing time, and (ii) a numerical estimation via discrete-time propagation under stochastic delay and corrupted velocity perception for tightened scenario evaluation. *Analytical bound (conservative):* If the attacker causes a perceived speed reduction such that the relative velocity  $\Delta v = v_i - \tilde{v}_{i-1} > 0$ , the time until the inter-vehicle gap closes is approximately  $s_i/\Delta v$ . If the detection latency  $t_{\text{det}}$  exceeds this time, a collision is deemed inevitable. Hence, this condition defines a binary outcome (collision or no collision), producing a conservative probabilistic bound:

$$P_{\text{coll}|A} \gtrsim \Pr\left(t_{\text{det}} \geq \frac{s_i}{\Delta v}\right) \cdot \text{TPR}. \quad (49)$$

This conservative bound is complemented by a Monte Carlo-based evaluation that incorporates stochastic communication delay, braking dynamics, and varying end-to-end latency to obtain a more realistic continuous estimate of  $P_{\text{coll}|A}$ .

4) *Aggregation to Fleet-Level Collision Likelihood and Flow Effects:* Let the attack arrival rate (per vehicle) be  $\lambda_A$  (attacks/sec) and consider the time horizon  $T$ . The expected number of attack attempts in a fleet of  $N$  vehicles is  $N\lambda_A T$ . Under independence, expected collisions in the horizon  $T$  are:

$$\mathbb{E}[N_{\text{coll}}] \approx N\lambda_A T \cdot P_{\text{eff}} \cdot P_{\text{coll}|A}. \quad (50)$$

Use the conservative bound for  $P_{\text{coll}|A}$  above when analytic tractability is required.

*Flow-level effect (Vehicle Flow Efficiency, VFE):* Let baseline flow (no attack) at link be  $q_0 = k_0 \cdot v_0$  (vehicles/hr), where  $k_0$  is density and  $v_0$  average speed. Each collision or critical event causes a local disruption of duration  $T_{\text{disr}}$  and reduces throughput by  $\delta q$  during disruption. Then the expected mean flow reduction over  $T$  is:

$$\Delta q \approx \frac{\mathbb{E}[N_{\text{coll}}] \cdot \delta q \cdot T_{\text{disr}}}{T}, \quad \text{and} \quad \text{VFE} = 1 - \frac{\Delta q}{q_0}. \quad (51)$$

For small  $\mathbb{E}[N_{\text{coll}}]$ , linearization yields VFE close to 1.

5) *IDS-Induced Message Loss and False-Positive Consequences:* Let the total messages in the horizon  $T$  be  $M$ , of which a fraction  $p_{\text{attack}}$  is malicious. Using IDS confusion terms:

- True positives (TP):  $\text{TP} \approx M p_{\text{attack}} \cdot \text{TPR}$
- False positives (FP):  $\text{FP} \approx M(1 - p_{\text{attack}}) \cdot \text{FPR}$

If IDS mitigation blocks messages flagged as malicious, the Safety Message Delivery Ratio (SMDR) for legitimate safety messages is:

$$\text{SMDR} = \frac{M(1 - p_{\text{attack}}) - \text{FP}}{M(1 - p_{\text{attack}})} = 1 - \text{FPR}. \quad (52)$$

(If mitigation causes delay instead of drop, model delay effects below.)

If mitigation introduces additional per-message processing delay  $d_{\text{proc}}$ , and dropped/flagged messages are retransmitted or suppressed, the effective average CAM latency becomes:

$$\bar{d}_{\text{cam}} = d_{\text{net}} + \text{FPR} \cdot d_{\text{proc}} \quad (53)$$

where  $d_{\text{net}}$  is base network latency. Increased CAM latency can increase TTC uncertainty; incorporate via inflated  $t_c$  or larger  $\epsilon$  in TTC.

6) *Summary: TIMM Input-Output Mappings:* Collectively, TIMM yields closed-form or computable mappings from IDS performance to transportation impact metrics:

- Extra travel distance during detection:  $\Delta d = v t_{\text{det}}$
- Probability an attack remains effective:  $P_{\text{eff}} \approx p_{\text{attack}}(1 - \text{TPR} \cdot F_t(t_c))$
- Expected collisions per time:  $\mathbb{E}[N_{\text{coll}}] \approx N \lambda_A P_{\text{eff}} P_{\text{coll}|A}$
- Safety Message Delivery Ratio:  $\text{SMDR} = 1 - \text{FPR}$  (if blocked messages are primary mitigation)
- Vehicle Flow Efficiency:  $\text{VFE} = 1 - \frac{\mathbb{E}[N_{\text{coll}}] \cdot \delta q \cdot T_{\text{disr}}}{q_0 T}$

All mappings are parameterized by scenario variables  $\{v, s^*, s_i[0], \Delta t\}$  and IDS variables  $\{\text{TPR}, \text{FPR}, \bar{t}_{\text{det}}, F_t(\cdot)\}$ . In Section IV, we define the evaluation metrics, and in Section V, we present parametric sweeps and results.

#### IV. EXPERIMENTAL DETAILS

This section outlines the datasets used in our evaluation, followed by the architectural and training configuration of TCNnet. All experiments were conducted on a standard workstation equipped with an Intel Core i9 CPU and 32 GB of RAM, using Python with TensorFlow, Keras, Pandas, and NumPy in a lightweight, resource-efficient setup suitable for deployment on edge vehicles.

##### A. Datasets Detail

We evaluate the proposed framework using two real-world benchmark datasets: the Automotive Ethernet Intrusion Dataset (AEID) [19] and the TOW dataset [21]. AEID contains port-mirrored 100BASE-T1 traffic from a vehicle-mounted camera in both driving and non-driving scenarios, with replay attacks generated from 36 recorded packets. TOW provides 100BASE-T1 traffic with five attack types: replay, frame injection, MAC flooding, PTP manipulation, and CAN DoS, with each packet labeled as normal or attack. To assess scalability, we evaluate TCNnet using multiple NTTE-generated traffic-image sizes ( $\Xi^{[32 \times 32]}$ ,  $\Xi^{[60 \times 60]}$ ,  $\Xi^{[116 \times 116]}$ ,  $\Xi^{[452 \times 452]}$ ). For the TOW dataset, we apply data augmentation consisting of centering, standard normalization, a  $20^\circ$  rotation, and width/height shifts of 0.2 to improve generalization.

In our experimental setup, we use these datasets to evaluate TCNnet under diverse network scenarios and image sizes. Tables I and II summarize the dataset compositions, including normal and attack traffic across conditions. To emulate realistic deployment, five edge vehicles participate in privacy-preserving collaborative learning, each training on a distinct data partition. This introduces natural distribution shifts across scenarios, attack types, and image resolutions, enabling a

TABLE I  
AEID DATASET DETAILS FOR DIFFERENT NETWORK SCENARIOS

| Network Scenario | Traffic Type  | Packets | Training | Testing |
|------------------|---------------|---------|----------|---------|
| Non-Driving      | Normal        | 369K    | 295K     | 74K     |
|                  | Replay attack | 81K     | 65K      | 16K     |
| Driving          | Normal        | 1.29M   | 1.03M    | 258K    |
|                  | Replay attack | 325K    | 260K     | 65K     |

K = 1,000 (thousand), M = 1,000,000 (million).

TABLE II  
TWO-IDS DATASET DETAILS FOR DIFFERENT TRAFFIC  
IMAGE SIZES AND ATTACK TYPES

| traffic image size       | Traffic Type       | Packets | Training | Testing |
|--------------------------|--------------------|---------|----------|---------|
| $\Xi^{[32 \times 32]}$   | Normal             | 1.16M   | 691K     | 472K    |
|                          | DoS attack         | 127K    | 85K      | 41K     |
|                          | CAN replay attack  | 60K     | 30K      | 30K     |
|                          | CAM table overflow | 51K     | 34K      | 17K     |
| $\Xi^{[60 \times 60]}$   | Normal             | 12K     | 7K       | 5K      |
|                          | PTP sync attack    | 91K     | 65K      | 26K     |
| $\Xi^{[116 \times 116]}$ | Normal             | 7.6K    | 4.4K     | 3.2K    |
|                          | DoS attack         | 4.3K    | 2.6K     | 1.6K    |
|                          | PTP sync attack    | 18K     | 9.6K     | 8.3K    |
|                          | Injection attack   | 10.5K   | 8K       | 2.6K    |
| $\Xi^{[452 \times 452]}$ | Normal             | 433K    | 252K     | 181K    |
|                          | Injection attack   | 52K     | 35K      | 17K     |

K = 1,000 (thousand), M = 1,000,000 (million).

realistic assessment of the detector's robustness and its ability to generalize to unseen traffic patterns.

##### B. TCNnet Architectural and Experimental Settings

TCNnet is designed as a lightweight intrusion detector for OVN, combining convolutional layers for feature extraction with layer normalization for stable training. We adopt a two-level learning strategy: edge vehicles train using the Adam optimizer with a learning rate of 0.01, while the aggregator updates the global model with a learning rate of 0.05. The number of communication rounds is adjusted per dataset and scenario to ensure stable convergence. Architectural details, hyperparameters, and training settings are summarized in Table IV.

##### C. Evaluation Metrics

1) *Standard IDS Performance Metrics:* We evaluate TCNnet using standard intrusion detection metrics: Accuracy (ACC), Precision (PRE), Recall (REC), Area Under the Curve (AUC), False Positive Rate (FPR), True Positive Rate (TPR), and detection latency ( $t_{\text{det}}$ ) [26].

2) *Transportation Impact Assessment:* To quantitatively evaluate the real-world safety and efficiency consequences of cyber-attacks and our IDS, we employ the proposed TIMM. This analytical framework establishes formal mappings from IDS performance metrics (TPR, FPR, latency) to critical transportation-level outcomes, including SMDR, collision likelihood, and VF, as detailed in Section III-D. The car-following dynamics and controller model within TIMM are grounded in established Adaptive Cruise Control research [22], [23]. The safety assessment is based on the formalized TTC metric, a recognized standard for collision risk analysis [24], while the overall cybersecurity-to-safety evaluation addresses a critical gap identified in connected vehicle security [25]. The specific

TABLE III  
PARAMETERS FOR TIMM-BASED INTRUSION IMPACT EVALUATION

| Parameter                    | Symbol / Value                                        | Description                      |
|------------------------------|-------------------------------------------------------|----------------------------------|
| <b>Global Parameters</b>     |                                                       |                                  |
| No. of vehicles              | $N = 1000$                                            | Fleet size in simulation         |
| Attack rate                  | $\lambda_A = 10^{-4}$                                 | Attacks per vehicle per hour     |
| Observation window           | $T = 3600$ s                                          | Simulation duration              |
| Disruption duration          | $T_{\text{disr}} = 300$ s                             | Time of flow disruption          |
| Flow degradation coeff.      | $\delta_q = 0.8$                                      | Throughput loss factor           |
| Nominal flow rate            | $q_0 = 1.0$                                           | Baseline normalized flow         |
| IDS internal delay           | $d_{\text{proc}} = 1$ ms                              | Processing delay per packet      |
| Critical TTC threshold       | $\tau_{\text{crit}} = 2.0$ s                          | Safety margin before collision   |
| Minimum collision gap        | $s_{\text{min}} = 2.0$ m                              | Distance for collision detection |
| System latency               | $t_{\text{sys}} \in [0, 5]$ ms                        | End-to-end detection latency     |
| <b>Scenario Parameters</b>   |                                                       |                                  |
| Urban speed                  | $v_{\text{urban}} = 30$ km/h                          | Intra-city traffic               |
| Suburban speed               | $v_{\text{suburban}} = 60$ km/h                       | Mixed-speed traffic              |
| Highway speed                | $v_{\text{highway}} = 100$ km/h                       | High-speed scenario              |
| Urban headway                | $s_{\text{star}} = 15$ m, $s_{\text{initial}} = 10$ m | Desired/initial spacing          |
| Suburban headway             | $s_{\text{star}} = 30$ m, $s_{\text{initial}} = 25$ m | Desired/initial spacing          |
| Highway headway              | $s_{\text{star}} = 50$ m, $s_{\text{initial}} = 45$ m | Desired/initial spacing          |
| Urban attack probability     | $p_A = 0.02$                                          | Fraction of compromised nodes    |
| Suburban attack probability  | $p_A = 0.05$                                          | Fraction of compromised nodes    |
| Highway attack probability   | $p_A = 0.10$                                          | Fraction of compromised nodes    |
| Urban spoofing factor        | $\alpha = 0.3$                                        | Closing-speed amplification      |
| Suburban spoofing factor     | $\alpha = 0.5$                                        | Closing-speed amplification      |
| Highway spoofing factor      | $\alpha = 0.7$                                        | Closing-speed amplification      |
| <b>Controller Parameters</b> |                                                       |                                  |
| Speed gain                   | $K_p = 0.8$                                           | Velocity tracking gain           |
| Headway gain                 | $K_h = 0.3$                                           | Spacing regulation gain          |
| Time step                    | $\Delta t = 0.1$ s                                    | Discrete simulation interval     |

Note: Parameter justifications: Vehicle dynamics [22], [23], safety thresholds [24], security context [25].

TABLE IV  
HYPERPARAMETERS AND IMPLEMENTATION DETAILS

| Hyperparameters      |                                                                                                                                                                                             |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conv Layers          | 32, 32                                                                                                                                                                                      |
| Layer Norm           | Yes                                                                                                                                                                                         |
| Pooling              | Max (2,2)                                                                                                                                                                                   |
| Dense Units          | 32                                                                                                                                                                                          |
| Activation           | ReLU, Sigmoid (AEID)<br>Softmax (TOW)                                                                                                                                                       |
| Edge-Level LR        | 0.01                                                                                                                                                                                        |
| Edge-Level Optimizer | Adam                                                                                                                                                                                        |
| Epochs               | 10                                                                                                                                                                                          |
| Agg. LR              | 0.05                                                                                                                                                                                        |
| Agg. Optimizer       | Adam                                                                                                                                                                                        |
| Loss                 | Binary CE (AEID)<br>Categorical CE (TOW)                                                                                                                                                    |
| Batch Size           | 16 (TOW)<br>32 (AEID)                                                                                                                                                                       |
| No. of Vehicles      | 10 (TOW)<br>5 (AEID)                                                                                                                                                                        |
| Comm. Rounds         | 100 (Non-Driving Network Scenario)<br>20 (Driving Network Scenario)<br>30 ( $\Xi_{[32 \times 32]}$ & $\Xi_{[60 \times 60]}$ )<br>50 ( $\Xi_{[116 \times 116]}$ & $\Xi_{[452 \times 452]}$ ) |

Conv Layers: Convolutional layers, Layer Norm: Layer normalization, Pooling: Max pooling operation. Edge LR: Learning rate for edge vehicles, Agg. LR: Learning rate for aggregator. CE: Cross-entropy, Comm. Rounds: Number of communication rounds.

parameter settings for the Urban, Suburban, and Highway scenarios are provided in Table III.

## V. PERFORMANCE EVALUATION

In this section, we assess the performance of the proposed framework. Our experimental objectives are threefold: first, to build, train, and evaluate the TCNnet within a

TABLE V  
PERFORMANCE EVALUATION OF TCNNET OVER MULTIPLE COMMUNICATION ROUNDS ON THE AEID DATASET FOR NON-DRIVING AND DRIVING SCENARIOS. THE RESULTS SHOWCASE ACC, PRE, REC, AUC, AND LOSS AT DIFFERENT ROUNDS OF THE LEARNING PROCESS

| Network Scenario | Metric | Communication Rounds |                 |                 |                 |                  |                  |                  |                  |                  |                  |
|------------------|--------|----------------------|-----------------|-----------------|-----------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                  |        | 2 <sup>nd</sup>      | 4 <sup>th</sup> | 6 <sup>th</sup> | 8 <sup>th</sup> | 10 <sup>th</sup> | 12 <sup>th</sup> | 14 <sup>th</sup> | 16 <sup>th</sup> | 18 <sup>th</sup> | 20 <sup>th</sup> |
| Driving          | ACC    | 93.44                | 97.02           | 99.08           | 99.77           | 99.96            | 99.97            | 99.95            | 99.95            | 99.97            | 99.98            |
|                  | PRE    | 88.15                | 92.81           | 98.08           | 99.64           | 99.93            | 99.93            | 99.86            | 99.93            | 99.93            | 99.99            |
|                  | REC    | 72.76                | 90.15           | 96.71           | 99.09           | 99.86            | 99.93            | 99.86            | 99.79            | 99.93            | 99.93            |
|                  | AUC    | 96.08                | 99.17           | 99.90           | 99.99           | 99.99            | 99.99            | 99.99            | 99.96            | 99.96            | 99.99            |
|                  | Loss   | 0.181                | 0.082           | 0.047           | 0.017           | 0.060            | 0.002            | 0.001            | 0.001            | 0.001            | 0.001            |
| Non-Driving      | ACC    | 85.36                | 92.03           | 95.77           | 97.87           | 98.50            | 99.22            | 99.50            | 99.77            | 99.84            | 99.86            |
|                  | PRE    | 76.85                | 87.31           | 93.53           | 96.53           | 97.51            | 98.79            | 99.15            | 99.68            | 99.77            | 99.73            |
|                  | REC    | 75.65                | 87.00           | 92.79           | 96.61           | 97.68            | 98.70            | 99.24            | 99.59            | 99.73            | 99.82            |
|                  | AUC    | 93.04                | 97.60           | 99.15           | 99.63           | 99.60            | 99.75            | 99.76            | 99.82            | 99.92            | 99.91            |
|                  | Loss   | 0.313                | 0.189           | 0.110           | 0.059           | 0.052            | 0.030            | 0.027            | 0.016            | 0.007            | 0.008            |

privacy-preserved collaborative learning framework. Second, to assess the resource efficiency of TCNnet from various aspects and its feasibility. Third, to conduct a comparative analysis against state-of-the-art methods.

### A. Detection Results of Proposed Framework

The experimental setup for TCNnet employs collaborative learning, where the aggregator initializes the model and edge vehicles, denoted as  $N$ , update it in a privacy-preserving manner by sharing only model parameters. For evaluation, we set  $N = 5$  for AEID and  $N = 10$  for TOW to cover diverse deployment scenarios.

For AEID, TCNnet converged in 100 rounds for the non-driving scenario due to complex, varied attack patterns, and in 20 rounds for the driving scenario with more structured attacks. Upon convergence, non-driving results were ACC 99.86%, PRE 99.73%, REC 99.82%, AUC 99.91%, and loss 0.008; driving results were ACC 99.98%, PRE 99.99%, REC 99.93%, AUC 99.99%, and loss 0.001. Detailed results across rounds are presented in Table V, confirming TCNnet's effectiveness in learning attack patterns while preserving privacy. TCNnet was evaluated across multiple NTTE-generated traffic image sizes to assess adaptability to diverse protocols. Sizes ranged from  $32 \times 32$  to  $452 \times 452$ . For  $32 \times 32$  and  $60 \times 60$ , learning was smooth, with convergence in 50 and 15 communication rounds, respectively. Accuracy reached 99.99%, with precision slightly lower at 99.95% for  $32 \times 32$ . For  $116 \times 116$ , initial fluctuations stabilized after 30 rounds, achieving ACC 99.99%, PRE 99.95%, REC 99.99%, and AUC 99.99%. The largest size,  $452 \times 452$ , showed significant early fluctuations (ACC 55% in the first 10 rounds) but eventually reached 99.99% across all metrics. Detailed results across rounds and image sizes are summarized in Table VI.

### B. Embedded-Optimized Quantized TCNnet

Designing a tiny intrusion detection model is only half the challenge; ensuring that it remains highly effective after quantization is equally important. To enable deployment in resource-constrained on-board automotive networks, we quantized TCNnet into Q-TCNnet using TensorFlow Lite, converting all FP32 weights into 8-bit INT8 representations through post-training quantization. Despite this substantial reduction in numerical precision, Q-TCNnet preserves performance extremely well: as summarized in Table VII, accuracy,

TABLE VI

PERFORMANCE EVALUATION OF TCNNET OVER MULTIPLE COMMUNICATION ROUNDS ON THE TOW DATASET IN A COLLABORATIVE LEARNING SETTING FOR DIFFERENT TRAFFIC IMAGE SIZES. THE TABLE PRESENTS ACC, PRE, REC, AUC, AND LOSS FOR EACH SIZE AT VARIOUS ROUNDS OF THE LEARNING PROCESS

| Traffic Image Size    | Metric | Communication Rounds |                  |                  |                  |                  |                  |                  |                  |                  |                  |
|-----------------------|--------|----------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
|                       |        | 5 <sup>th</sup>      | 10 <sup>th</sup> | 15 <sup>th</sup> | 20 <sup>th</sup> | 25 <sup>th</sup> | 30 <sup>th</sup> | 35 <sup>th</sup> | 40 <sup>th</sup> | 45 <sup>th</sup> | 50 <sup>th</sup> |
| $\Xi[32 \times 32]$   | ACC    | 99.73                | 99.85            | 99.90            | 99.91            | 99.95            | 99.96            | 99.95            | 99.97            | 99.97            | 99.99            |
|                       | PRE    | 98.97                | 99.43            | 99.64            | 99.67            | 99.73            | 99.77            | 99.85            | 99.89            | 99.92            | 99.95            |
|                       | REC    | 99.52                | 99.72            | 99.83            | 99.84            | 99.90            | 99.92            | 99.95            | 99.95            | 99.97            | 99.99            |
|                       | AUC    | 99.92                | 99.89            | 99.93            | 99.96            | 99.94            | 99.95            | 99.98            | 99.95            | 99.98            | 99.99            |
|                       | Loss   | 0.010                | 0.007            | 0.006            | 0.004            | 0.003            | 0.003            | 0.002            | 0.002            | 0.001            | 0.001            |
| $\Xi[60 \times 60]$   | ACC    | 91.82                | 99.15            | 99.82            | 99.96            | 99.99            | 99.98            | 99.98            | 99.98            | 99.98            | 99.99            |
|                       | PRE    | 92.26                | 99.30            | 99.87            | 99.98            | 99.99            | 99.99            | 99.99            | 99.98            | 99.99            | 99.99            |
|                       | REC    | 99.17                | 99.75            | 99.92            | 99.98            | 99.99            | 99.99            | 99.99            | 99.98            | 99.99            | 99.99            |
|                       | AUC    | 85.53                | 99.41            | 99.86            | 99.99            | 99.99            | 99.95            | 99.95            | 99.96            | 99.96            | 99.99            |
|                       | Loss   | 0.233                | 0.031            | 0.007            | 0.001            | 0.001            | 0.001            | 0.001            | 0.001            | 0.001            | 0.001            |
| $\Xi[116 \times 116]$ | ACC    | 98.37                | 99.08            | 99.51            | 99.62            | 99.88            | 99.85            | 99.95            | 99.98            | 99.99            | 99.99            |
|                       | PRE    | 97.05                | 99.06            | 99.44            | 99.61            | 99.83            | 99.83            | 99.88            | 99.96            | 99.96            | 99.95            |
|                       | REC    | 99.88                | 99.17            | 99.61            | 99.66            | 99.94            | 99.88            | 99.97            | 99.99            | 99.99            | 99.99            |
|                       | AUC    | 99.66                | 99.97            | 99.99            | 99.96            | 99.94            | 99.99            | 99.99            | 99.99            | 99.99            | 99.99            |
|                       | Loss   | 0.052                | 0.024            | 0.014            | 0.011            | 0.006            | 0.005            | 0.004            | 0.002            | 0.001            | 0.001            |
| $\Xi[452 \times 452]$ | ACC    | 50.90                | 56.26            | 56.96            | 81.61            | 95.95            | 99.59            | 99.97            | 99.99            | 99.98            | 99.99            |
|                       | PRE    | 50.93                | 55.73            | 55.82            | 79.58            | 94.37            | 97.64            | 99.68            | 99.96            | 99.99            | 99.99            |
|                       | REC    | 65.86                | 75.10            | 63.65            | 86.94            | 98.79            | 97.79            | 99.79            | 99.91            | 99.99            | 99.99            |
|                       | AUC    | 50.61                | 55.53            | 60.53            | 92.23            | 98.95            | 99.74            | 99.99            | 99.99            | 99.99            | 99.99            |
|                       | Loss   | 0.723                | 0.694            | 0.715            | 0.433            | 0.256            | 0.117            | 0.008            | 0.002            | 0.0001           | 0.0001           |

TABLE VII

PERFORMANCE AND RESOURCE COMPARISON OF TCNNET AND Q-TCNNET

| Metric            | TCNnet | Q-TCNnet | $\Delta$ (Q - T) |
|-------------------|--------|----------|------------------|
| Accuracy (%)      | 99.96  | 99.92    | -0.04            |
| Precision (%)     | 99.92  | 99.90    | -0.02            |
| Recall (%)        | 99.93  | 99.90    | -0.03            |
| AUC (%)           | 99.98  | 99.95    | -0.03            |
| Storage Size (MB) | 2.81   | 0.23     | -2.58            |
| Memory Usage (MB) | 0.91   | 0.44     | -0.47            |

precision, recall, and AUC decrease by only 0.04% or less. Conversely, quantization yields major resource savings. Storage requirements drop by more than 90% (2.81 MB to 0.23 MB), and memory usage is reduced by over 51% (0.91 MB to 0.44 MB). These reductions make Q-TCNnet highly suitable for real-time, embedded deployment within on-board automotive networks where computational and memory budgets are tightly constrained.

### C. Comparison With Existing SOTA Methods

We evaluated our proposed IDS by comparing its performance against state-of-the-art (SOTA) methods. The evaluation was based on key performance metrics, including ACC, PRE, REC, and AUC. These comparative results were taken directly from the literature to maintain consistency with the original studies and ensure transparency, providing readers with an accurate reference for performance benchmarking. For the AEID dataset, we conducted a comprehensive performance comparison against SOTA methods, including those presented in [16], [19], [27], and [28]. For the first three SOTA methods [16], [27], [28], we were able to compare performance only within the driving scenario due to the unavailability of non-driving network scenario results. However, for the fourth method [19], we compared performance in both scenarios using all key metrics.

In the driving scenario, our proposed TCNnet and its quantized version (Q-TCNnet) outperformed the most recent multi-stage method [16]. TCNnet achieved an ACC of 99.99%, while Q-TCNnet obtained 99.96%, both surpassing the 97.97%

TABLE VIII

COMPARISON OF DIFFERENT IDS MODELS ON AEID AND TOW DATASETS

| Dataset | Scenario              | Model             | ACC   | PRE   | REC   | AUC   |
|---------|-----------------------|-------------------|-------|-------|-------|-------|
| AEID    | Driving               | Proposed TCNnet   | 99.99 | 99.98 | 99.95 | 99.99 |
|         |                       | Proposed Q-TCNnet | 99.96 | 99.97 | 99.94 | 99.98 |
|         |                       | Multi-Stage [16]  | 97.97 | 97.95 | 97.97 | 97.89 |
|         |                       | P-2DCNN [27]      | 96.13 | 96.98 | 97.84 | 96.74 |
|         |                       | XGBoost [28]      | 95.74 | 95.27 | 93.57 | 94.05 |
|         | Non-Driving           | V-2DCNN [19]      | 97.19 | 96.37 | 95.79 | 94.89 |
|         |                       | Proposed TCNnet   | 99.84 | 99.74 | 99.84 | 99.93 |
|         |                       | Proposed Q-TCNnet | 99.82 | 99.74 | 99.82 | 99.91 |
|         |                       | V-2DCNN [19]      | 96.55 | 98.13 | 98.40 | 98.97 |
|         |                       | Proposed TCNnet   | 99.98 | 99.99 | 99.97 | 99.99 |
| TOW     | $\Xi[452 \times 452]$ | Proposed Q-TCNnet | 99.96 | 99.97 | 99.93 | 99.95 |
|         |                       | Multi-Stage [16]  | 97.62 | 96.60 | 97.62 | 97.33 |
|         |                       | TOW-based [21]    | 97.87 | 95.80 | 97.90 | 94.33 |
|         |                       | FASS [29]         | 97.00 | 94.00 | 95.00 | 96.33 |
|         | $\Xi[116 \times 116]$ | Proposed TCNnet   | 99.98 | 99.93 | 99.97 | 99.99 |
|         |                       | Proposed Q-TCNnet | 99.94 | 99.89 | 99.91 | 99.95 |
|         |                       | TOW-based [21]    | 96.11 | 97.55 | 95.55 | 97.53 |
|         | $\Xi[60 \times 60]$   | Proposed TCNnet   | 99.97 | 99.98 | 99.94 | 99.99 |
|         |                       | Proposed Q-TCNnet | 99.92 | 99.95 | 99.90 | 99.95 |
|         |                       | TOW-based [21]    | 97.39 | 96.35 | 95.22 | 98.23 |
|         | $\Xi[32 \times 32]$   | Proposed TCNnet   | 99.98 | 99.91 | 99.89 | 99.98 |
|         |                       | Proposed Q-TCNnet | 99.94 | 99.87 | 99.88 | 99.93 |
|         |                       | TOW-based [21]    | 92.32 | 90.55 | 91.88 | 92.35 |

ACC of the multi-stage IDS by over 2%. Moreover, our models demonstrated superior AUC scores, with TCNnet achieving 99.99% and Q-TCNnet 99.98%, compared to 97.89% for the multi-stage approach. These results were obtained despite running only 10 epochs at the edge vehicle level, considering resource constraints in real-world environments.

In the network scenario, our TCNnet and Q-TCNnet models performed exceptionally well, significantly outperforming the V-2DCNN [19] across all performance metrics. TCNnet achieved an ACC of 99.84% and Q-TCNnet 99.82%, compared to 96.55% for V-2DCNN. Furthermore, our collaborative learning framework provides an adaptive advantage, allowing edge vehicles to share attack patterns in a privacy-preserving manner, enhancing detection capabilities over time.

For the TOW dataset, both TCNnet and Q-TCNnet performed exceptionally well across all traffic image sizes. For the  $\Xi[32 \times 32]$  image size, our models achieved ACC scores of 99.98% and 99.94%, respectively, significantly outperforming the 92.32% of the TOW-based IDS by over 7.5%. Similarly, for  $\Xi[60 \times 60]$ , our models achieved a 2.58% and 2.53% higher ACC than the TOW-based approach. For  $\Xi[116 \times 116]$ , both models outperformed the baseline by over 3.8%. At the  $\Xi[452 \times 452]$  size, our models achieved 99.98% and 99.96% ACC, outperforming the multi-stage IDS (97.62%) and the TOW-based approach (97.87%) by more than 2%. The FASS method also performed significantly lower; however, its semi-supervised nature provides a different learning advantage. Detailed comparison results for both datasets can be found in Table VIII.

These results highlight the robustness of our collaborative learning framework, demonstrating its ability to detect and adapt to new attack patterns effectively in resource-constrained environments.

### D. ROC and Latency Evaluation of TCNnet

To evaluate the operational sensitivity of the proposed TCNnet detector, we computed the TPR, FPR, and inference latency per packet size across multiple decision thresholds. As summarized in Table IX, TCNnet maintains near-zero



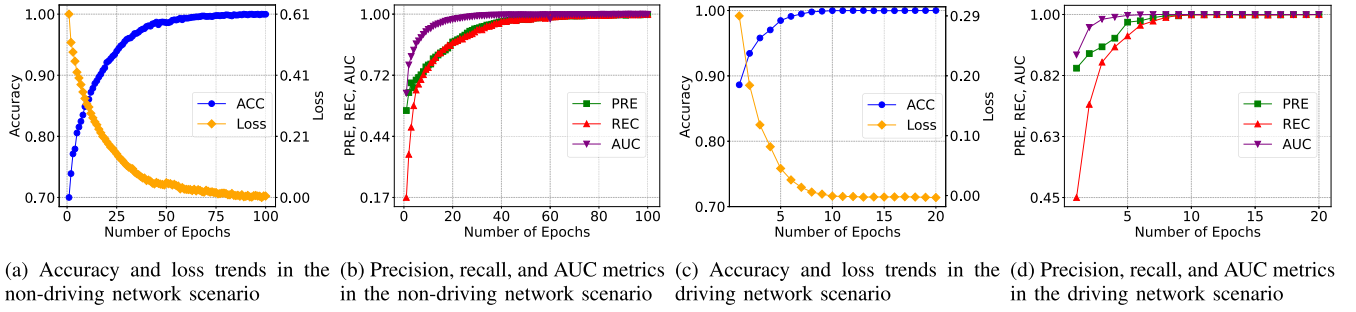


Fig. 5. Performance evaluation of the proposed TCNnet on the AEID dataset across different network scenarios.

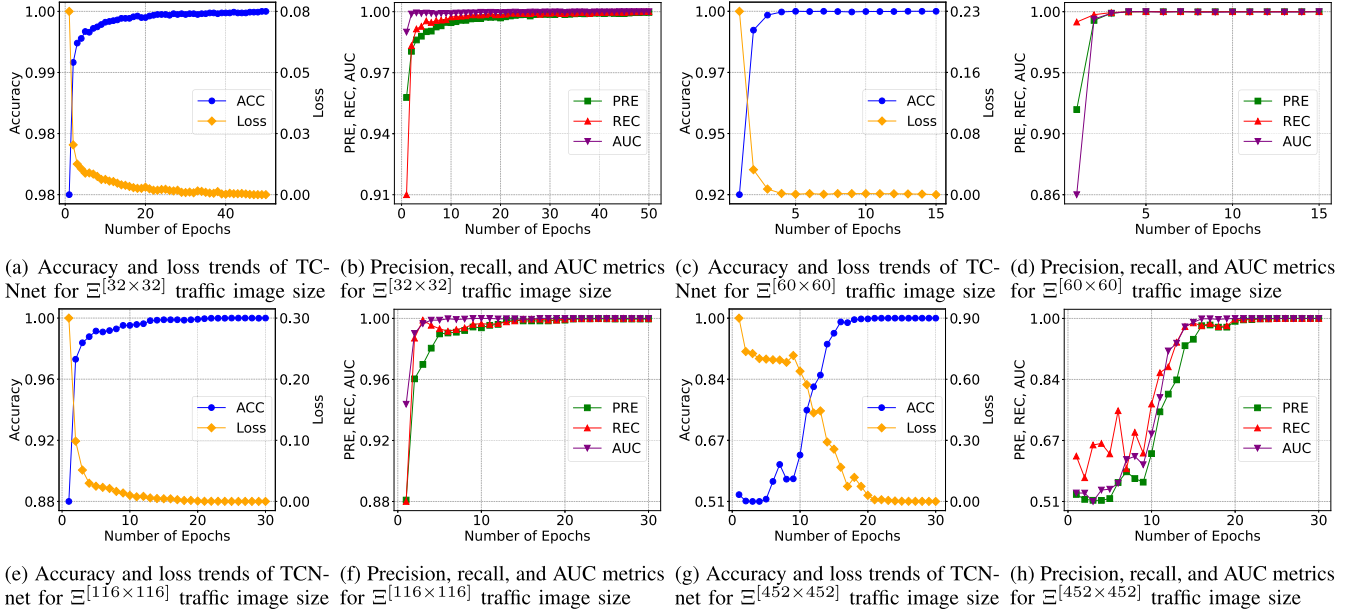


Fig. 6. Performance evaluation of the proposed TCNnet on the TOW dataset across different traffic image sizes.

TABLE IX  
ROC POINTS OF THE PROPOSED TCNNET DETECTOR  
AND AVERAGE LATENCY

| Threshold       | FPR        | TPR        |
|-----------------|------------|------------|
| 0.99999022      | 0.00044647 | 0.64714315 |
| 0.99997914      | 0.00053576 | 0.71042392 |
| 0.99996436      | 0.00062506 | 0.75138235 |
| 0.99993765      | 0.00080364 | 0.78988327 |
| 0.99970907      | 0.00116082 | 0.85050174 |
| 0.98993057      | 0.00437539 | 0.93938153 |
| 0.92453676      | 0.00884008 | 0.96743805 |
| 0.66726929      | 0.01446558 | 0.98402621 |
| 0.16262688      | 0.02946692 | 0.99344665 |
| 0.00000000      | 0.78337352 | 1.00000000 |
| Average Latency | 0.086 ms   |            |

false alarms (FPR < 0.001) at high thresholds while gradually improving TPR from 64.7% to 98.4% as the threshold is relaxed. This behavior reflects a stable and discriminative model, capable of maintaining detection reliability even under varying operating points. The consistent latency of 0.086 ms demonstrates TCNnet's feasibility for real-time ITS deployment, where ultra-low delay is essential for safety-critical decision loops. These results represent the *average performance of TCNnet across both AEID and TOW datasets*,

covering diverse packet sizes and network scenarios. Such cross-domain consistency highlights the robustness of the collaborative learning setup and provides a realistic view of detector behavior under different ROC operating points. These baseline results are critical for assessing the practicality of the proposed IDS in intelligent transportation systems, and serve as the foundation for subsequent subsections analyzing its network-level impact through TIMM-based simulations.

### E. TIMM-Based Evaluation of the Proposed IDS

To assess the effectiveness of the proposed IDS in realistic vehicular conditions, we performed extensive simulations using the TIMM framework across three representative traffic scenarios: *Urban*, *Suburban*, and *Highway*. The analysis considered  $N = 1000$  vehicles over a 1-hour window ( $T = 3600$  s) with detailed parameters provided in Table III.

Fig. 8 and Fig. 7 present the core relationship between IDS performance metrics and safety outcomes across all ROC operating points derived from our model validation (Table IX). Fig. 8 shows that collisions increase gradually with higher false positive rates, particularly in Highway scenarios where the impact is most pronounced due to higher speeds and attack probabilities. Conversely, Fig. 7 demonstrates the

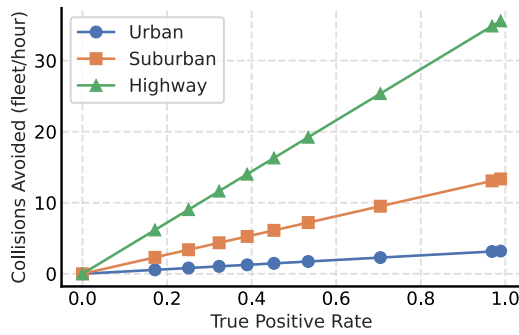


Fig. 7. Collisions avoided vs TPR at ROC Points for different traffic scenarios.

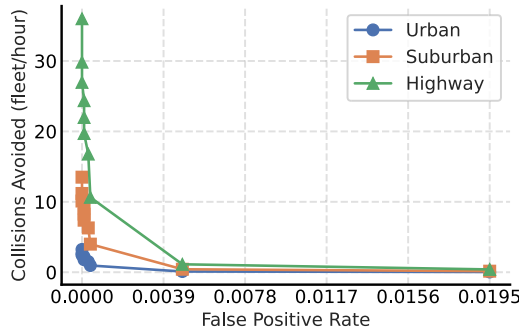


Fig. 8. Collisions avoided vs FPR at ROC points for different traffic scenarios.

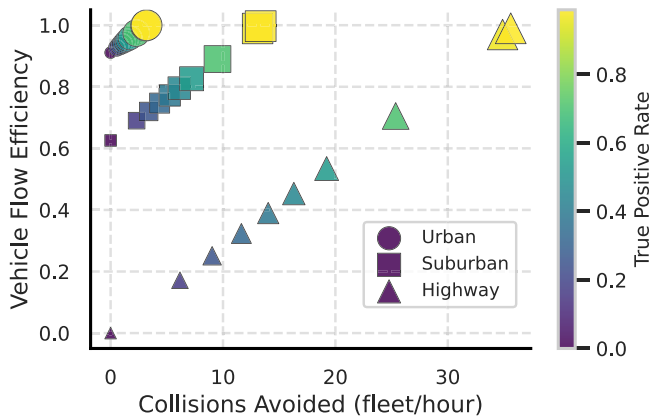


Fig. 9. Safety-Efficiency Trade-off: Vehicle Flow Efficiency (VFE) vs. Collisions Avoided across ROC operating points. Higher TPR values achieve better safety without compromising efficiency.

critical importance of detection sensitivity, with collision counts dropping sharply as TPR approaches 0.95 across all scenarios, highlighting the threshold beyond which substantial safety benefits are achieved. Moreover, the fundamental safety-efficiency trade-off is further examined in Fig. 9, which reveals that optimal operating points (TPR > 0.95) simultaneously maximize both collision avoidance and traffic flow efficiency. This demonstrates that our IDS achieves safety improvements without the typical efficiency penalties observed in conservative security systems. The color gradient (TPR) shows that higher detection sensitivity correlates with superior performance across both safety and efficiency metrics.

TABLE X

SCENARIO-LEVEL SUMMARY OF IDS PERFORMANCE AT ROC POINTS. THE COLLISION VALUES AND REDUCTION PERCENTAGES INCLUDE  $\pm$  VARIATION FROM MONTE CARLO SIMULATIONS WITH 1000 REPETITIONS

| Scenario | Coll. w/o IDS | Coll. w/ IDS    | Coll. Red (%)  | VFE / SMDR    |
|----------|---------------|-----------------|----------------|---------------|
| Highway  | 36.00         | $0.76 \pm 0.01$ | $97.9 \pm 1.2$ | 0.949 / 0.988 |
| Suburban | 13.50         | $0.29 \pm 0.01$ | $97.9 \pm 1.1$ | 0.981 / 0.991 |
| Urban    | 3.24          | $0.07 \pm 0.01$ | $97.9 \pm 1.0$ | 0.995 / 0.994 |

Note: Coll. w/o IDS = Collisions without IDS, Coll. w/IDS = Collisions with IDS, Coll. Red = Collision Reduction (%).

TABLE XI

PER-SCENARIO SUMMARY OF IDS PERFORMANCE BY SWEEP TYPE

| Scenario | Sweep Type           | Coll. Avoided (mean) | Coll. Avoided (min) | Coll. Avoided (max) | VFE (mean) | SMDR (mean) |
|----------|----------------------|----------------------|---------------------|---------------------|------------|-------------|
| Highway  | Expanded FPR Sweep   | 34.20                | 32.40               | 36.00               | 0.880      | 0.900       |
|          | ROC Operating Points | 35.24                | 34.87               | 35.60               | 0.949      | 0.988       |
|          | TNR-Based Sweep      | 34.20                | 32.40               | 36.00               | 0.880      | 0.950       |
| Suburban | Expanded FPR Sweep   | 12.82                | 12.15               | 13.50               | 0.955      | 0.960       |
|          | ROC Operating Points | 13.21                | 13.08               | 13.35               | 0.981      | 0.988       |
|          | TNR-Based Sweep      | 12.82                | 12.15               | 13.50               | 0.955      | 0.950       |
| Urban    | Expanded FPR Sweep   | 3.08                 | 2.92                | 3.24                | 0.989      | 0.900       |
|          | ROC Operating Points | 3.17                 | 3.14                | 3.20                | 0.995      | 0.988       |
|          | TNR-Based Sweep      | 3.08                 | 2.92                | 3.24                | 0.989      | 0.950       |

As shown in Table X, when evaluated at the top-performing ROC points (average TPR = 0.98, FPR = 0.0122, latency = 0.1 ms), the proposed IDS achieves remarkable safety improvements. In *Highway traffic*, the system reduces expected collisions from 36 to 0.76 per hour, corresponding to a  $97.9\% \pm 1.2\%$  reduction while maintaining high Vehicle Flow Efficiency (VFE = 0.949) and Safety Message Delivery Ratio (SMDR = 0.988). *Suburban conditions* show equally strong performance with collisions decreasing from 13.5 to 0.29 ( $97.9\% \pm 1.1\%$  reduction), VFE of 0.981, and SMDR of 0.991. *Urban traffic* benefits most in relative terms, with collisions dropping from 3.24 to 0.07 ( $97.9\% \pm 1.0\%$  reduction) and near-perfect VFE (0.995) and SMDR (0.994). These results, obtained from Monte Carlo simulations with 1000 repetitions, demonstrate consistent high-performance operation. These observed safety improvements are consistent with the probabilistic collision model  $P_{\text{coll}|A}$ , confirming that higher TPR and lower detection latency directly reduce the probability of collisions under realistic traffic scenarios.

To further examine the robustness of our approach, we conducted sensitivity analyses through expanded parameter sweeps summarized in Table XI. The *expanded FPR sweep* (FPR 0-0.20) reveals that even under artificially high false positive conditions, collision avoidance remains substantial (32.40-36.00 for Highway), demonstrating system resilience. The *TNR-based sweep* shows similar robustness, while the *ROC operating points* analysis confirms that proper threshold selection is crucial for maximizing safety benefits, with performance varying significantly across the ROC curve (collision avoidance range: 34.87–35.60 for Highway). Across all sweep types, the IDS maintains excellent VFE (> 0.88) and SMDR (> 0.90), confirming that traffic efficiency and communication reliability are preserved even under suboptimal operating conditions.

## VI. CONCLUSION

This work presented a privacy-preserving, embedded-efficient collaborative IDS framework for securing hybrid OANs. The NTTE module transforms raw multi-protocol

traffic into temporal images, while TCNnet and its quantized variant Q-TCNnet provide lightweight, high-performance intrusion detection suitable for resource-constrained ECUs. Post-training quantization reduced model size by over 90% and memory usage by 51% with negligible accuracy loss. Evaluations on AEID and TOW datasets confirmed that both models outperform state-of-the-art baselines. To bridge network security and real-world safety, we introduced TIMM, which maps IDS performance to ITS-level outcomes. Simulations across Urban, Suburban, and Highway settings showed up to 97.9% collision reduction, high VFE (0.949–0.995), and reliable SMDR (0.988–0.994), demonstrating tangible safety benefits. Despite these strengths, protocol-specific anomalies, rare attack types, and full integration with in-vehicle real-time systems remain open challenges. Future work includes protocol-aware embeddings, generative data augmentation, and deployment-oriented optimizations for ECUs and automotive gateways.

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